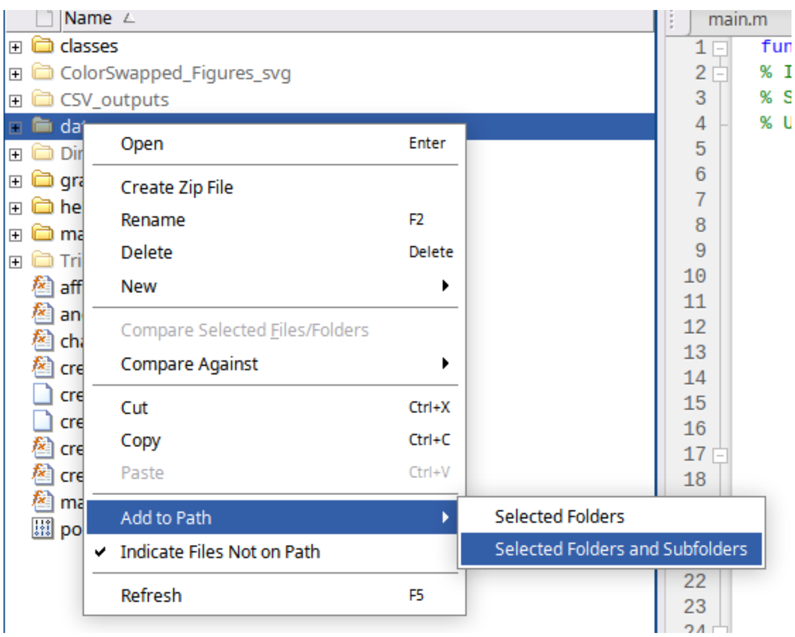
Using the Polymerization Toolkit

1. Copy the scripts. (This can be done by going to the Github, clicking the Code option, and downloading as ZIP). The directory will include:
    a. main.m
    b. /main_pipeline
    c. /helper_functions
    d. /graphing_and_analysis_scripts
    e. /classes
2. Prepare your table for analysis
    a. Cleaned data set, bin2. Refer to documentation for column headers if your data is not formatted in a Dynamo .tbl
    b. Note your bin2 pixel size
    c. Save the table to the /data directory in your scripts directory
3. Launch matlab and navigate to the scripts directory you just preprared
4. Right click on all folders and add each to Path



5. Call the main function by typing the following: <data_name>=main()
    a. The name should be descriptive enough to distinguish the data from others. Using the example table, we called this WT_CB as it's the wild type Rubisco in carboxysome dataset.

```
>> WT_CB=main()
```

6. The first input: "Is your input in a single file, or is it separated into even and odd files? Enter 1 for a single file or 2 for split files:"
    a. If you have the data divided into half sets this is 2. If you only have one table enter 1
    b. The example table is one table and enter 1 if you're using this to learn the toolkit

7. Next input is the name of the .tbl file to use in your data directory: "Enter the name of your data file. Please ensure that your input data is in bin1 or 2:"

```
fx Enter the name of your data file. Please ensure that your input data is in bin1 or 2: wildType.tbl
```

8. Now give the output file a name that is descriptive: "What would you like to call your processed input data file? Please end the name with .tbl:"

```
fx What would you like to call your processed input data file? Please end the name with .tbl: WT_out.tbl
```

9. Next, you need to provide the clean size in pixels: "What would you like the clean size to be?" This will depend on your project. This is the diameter in pixels that will be the threshold to remove overlapping particles. This is determined by biochemistry and the STA results; however, the general rule is to use .5 to .75 of the particle diameter. The pixel size here should be in bin2 if your data is bin2.
    a. For the example dataset, Rubisco is ~100Å in diameter. The bin2 pixel size is 2.14 Å/pix. Using a clean parameter of 32 we are removing particles that are within ~68Å.
    b. To refine this number to ensure this step isn't deleting real particles, one can run an alignment on the cleaned table. If the resolution decreases compared to the full dataset, that's an indication that real particles were removed and a less stringent clean value should be used.
10. You can run the scripts on bin1 or bin2 data but it will bin by 2 if you provide bin1. You will need to enter which bin size your data is currently in: "Is your input file in bin2? Enter 1 if your data is in bin1 or 2 if your data is bin2:"

```
fx Is your input file in bin2? Enter 1 if your data is in bin1 or 2 if your data is bin2: 2
```

11. Next, you need to provide your pixel size in meters: "Enter the pixel size, in meters, that your data uses. Please use scientific notation (ex. 2.54e-10) and keep in bin1 if you indicated that your input data was bin1:" (note: the bin2 pixel size for the example dataset is 2.14e-10)

```
fx s. Please use scientific notation (ex. 2.54e-10) and keep in bin1 if you indicated that your input data was bin1: 2.14e-10
```

12. Now, enter the diameter of the particle in meters: "Enter the diameter, in meters, of your particle. Please use scientific notation again:". For rubisco we use 9.936 nm

```
fx Enter the diameter, in meters, of your particle. Please use scientific notation again: 9.936e-9
```

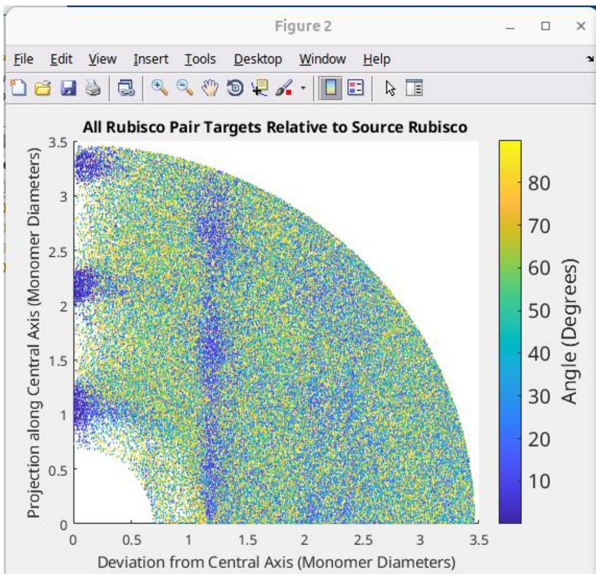
13. The processing will take a few minutes depending on the data size.
14. The program will now ask for a distance in which an interface is possible. This will vary on the protein: "Enter the maximum distance (in pixels) around a particle you want to search for potential oligomerization partners (to use default value of 2 * monomer diameter enter "default"):". For rubsico we use the default which is 9.936 nm. If you perfer to input your own number you should calculate the value, e.g. 1.5*9.936=14.904.

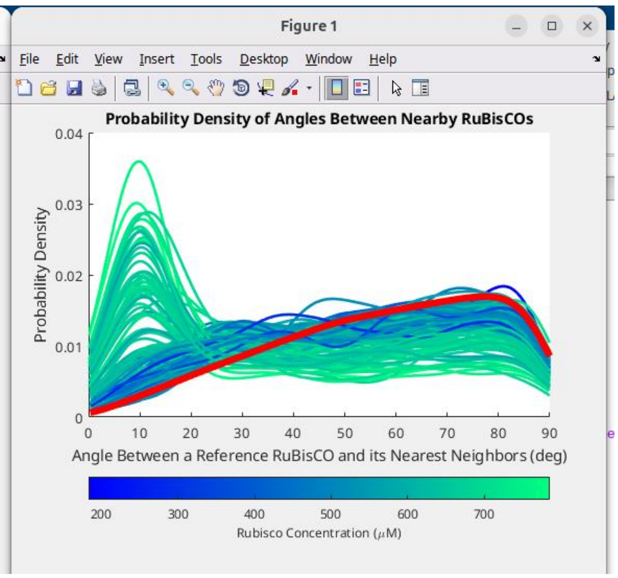
```
fx: to search for potential oligimerization partners (to use default value of 2 * monomer diameter enter "default"): default
```

15. The scripts will generate a series of plots to help determine parameters for additional processing, but you need to provide a value for the plot scale:
    "Linkage Analysis: Please use the following plots to help choose parameters for a valid linkage. Please enter the bandwidth for the plot (recommended value is 5):" We use the recommended value. This value will change the scale of the axis values for the visualization.

```
fx Please enter the bandwidth for the plot (recommended value is 5): 5
```

16. **Linkage analysis.** Two plots will be generated. This can be used to determine patterns in distances and angles in the linkages

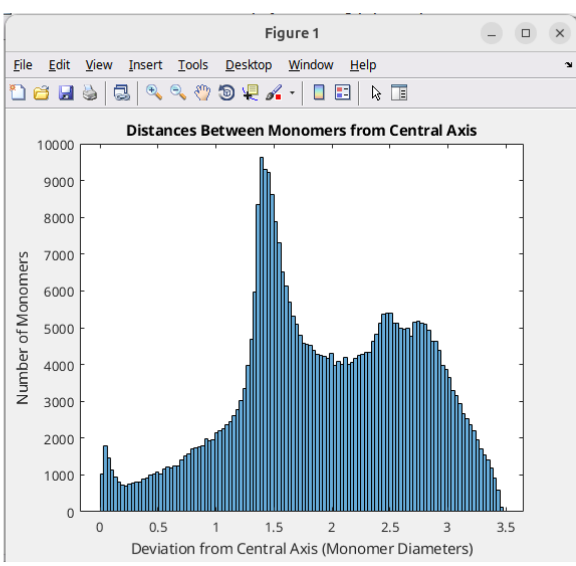


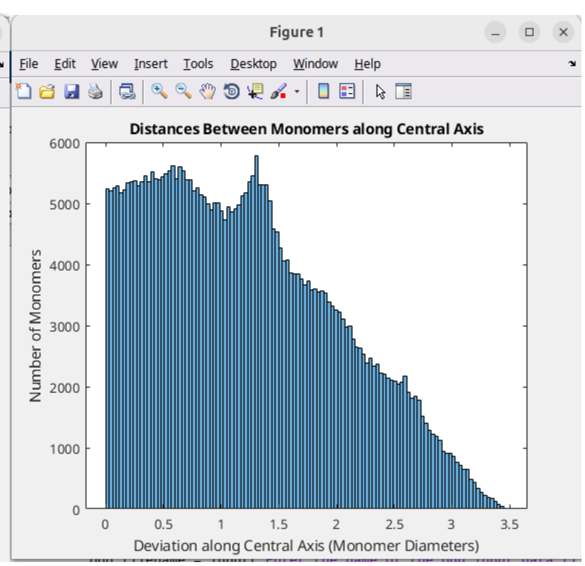


17. You will be asked in the next two questions if you want to adjust your y-axis to limit data. Use "inf" for your first round to see what the full dataset looks like before adjusting: "Enter the maximum y-axis value a data point can have in this plot (use "inf" to include all):" and "Enter the maximum x-axis value a data point can have in this plot (use "inf" to include all):"

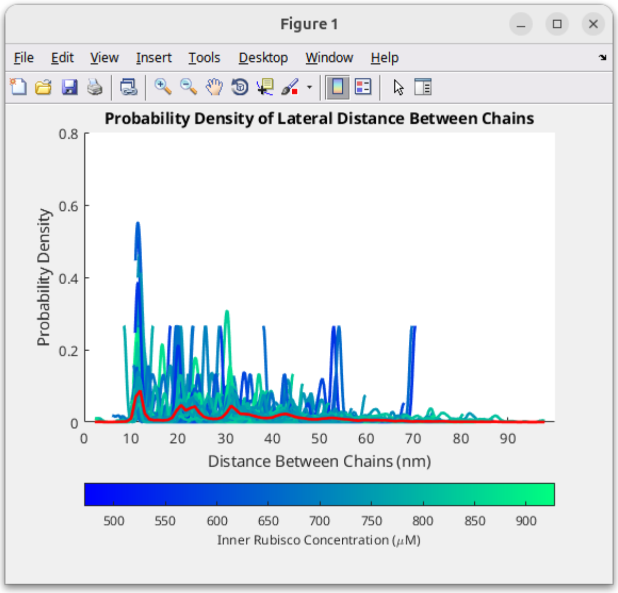
```
fx Enter the maximum y-axis value a data point can have in this plot (use "inf" to include all): inf
```

```
fx Enter the maximum x-axis value a data point can have in this plot (use "inf" to include all): inf
```

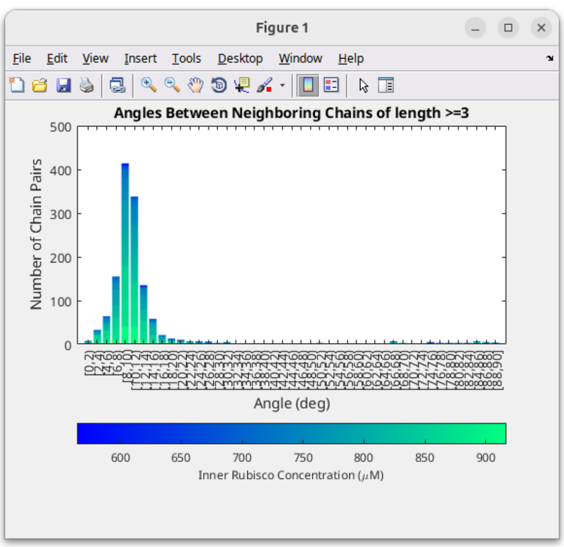




18. Building the linkage will ask if you want to customize any parameters based on the graphs. Using default for your first run is likely a good idea. “Would you like to use a custom set of parameters or a default set? Enter "custom" or "default":”
    a. Custom will ask you to define the minimum and maximum distance to search in terms of the particle diameter along the axis and radially from the axis.
    b. Custom will also as for the maximum angle between particles in degrees
    c. Custom will also ask if you want pairs to be in each other’s search volumes while building linkages. When set to true, this will check that the pairs are also in the same filament project. This is an extra check to ensure a filament is formed from the individual linkages.
19. Binding parameters. There are two different distances that can be used “tight” or “pivot”
    a. "Tight" parameters are more restricting, at 0.31 of a rubisco diameter for the radius_factor, and 25 degrees for the max_angle.
    b. "Pivot" parameters are more inclusive, at 0.45 of a rubisco diameter for the radius_factor and 50 degrees for the max_angle. All these recommended parameter values were determined from histogram analysis on our datasets.
    c. For Rubisco, we determined that 0.45 of a rubisco diameter is the maximum radius_factor that can be used in order to prevent multiple bindings. If a larger radius is used, the search might result in a source Rubisco having multiple top or bottom binding partners that satisfy the linkage conditions.
    d. Linkages are created with the user's choice for the set of parameters. We use tight for the rubisco polymerization. “Enter 0 to use "Tight" binding parameters, or 1 to use "Pivot" binding parameters:”
20. Chain Maker. Next you need to define the minimum number of particles that are required for an oligomer to be considered a chain. We use 3 for rubisco as the default, which can be selected by entering “default”.. “Enter the minimum length to be considered a chain or "default" to use the default value of 3:”
21. If there are more than one particle that could be bound to the same interface, you’ll get error messages here. There is a script that can be used to delete the particle that is least likely to be part of the interface and generate a cleaned table.
    a. If there are no error messages select no: “Have you received error messages that you would like to correct? (y/n):”
    b. If there are error messages select yes and the script will end with a new table of cleaned particles for processing. Simply restart the procedure using the new table from Step 5.
22. The scripts will analyze the data and recommend a value for the bandwidth: “Please enter the bandwidth for the plot (recommended value is 0.5):” Use that value



23. Using the graph, determine the best distance allowable between linked chains. We use 14 nm for rubisco: “Enter the maximum distance to allow between linked chains (in nm):”. Likewise, for minimum distance between chains we use 10nm.



24. For the chain linkages, there are custom parameters: *min_distance*, *max_distance*, and *max_angle* between two chains required for a set of chains to be considered a lattice. or use default. By default, *min_distance* will be set to 10 nm, *max_distance* will be set to 14nm, and *max_angle* will be set to 22 degrees. We use default for rubisco. It will then count the chain linkages and report. “Chain Linkages: Would you like to use a custom set of parameters or the default set? Enter "custom" or "default": default”
25. For the lattice, this will be determined by chain linkages. We use default for Rubisco. “Lattice Gen: Enter the minimum number of linked chains to constitute a lattice or "default" to use the default value of 5:”